

Salami ate my Neighbors!

Individual Project Design Documentation

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Abstract—This is a complete design documentation of the process behind the design lifecycle engaged in for the purposes of redesigning the user interface of *Salami ate my Neighbors!* by JKClayton.

1 INTRODUCTION

I am developing a stronger interface for the video game *Salami ate my Neighbors!* by JKClayton. This is a first-person shooter action game with a round-based, horde survival structure, inspired by series such as Call of Duty: Zombies and Plants vs. Zombies. While the game has been released publicly for over four years, the user interface is limited in terms of communicating vital information to the player.

The users playing the game have a seemingly simple, overarching goal: survive as long as possible. Stretched over the entirety of the game, there are many smaller goals that arise in the moment-to-moment gameplay. Players must scan the environment and their available resources to make calculated decisions that better their chances of survival. Much of this information is communicated to players through the user interface, namely the heads-up display (HUD). The current interface, however, has some glaring weaknesses that could be improved to better inform the user.

I have been involved with game development projects both individually and collaboratively over the past several years. Those collaborations only came to fruition thanks to Jackson (JKClayton), a friend from high school— the only person who seemed to be more fascinated by the inner workings of video games than I was. While he was focused on the programming and overall design of his games, I was routinely offered the chance to test builds or provide assistance in

asset creation. Several years ago, Jackson invited me to collaborate on a game jam project where I was tasked with creating the UI. While my designs were motivated by purely stylistic means, it made me realize that the way in which information is conveyed matters much more than how it's stylized. Since then, I've tried my hand at mirroring and understanding the process behind the design and implementation of video game UI.

2 NEEDFINDING PLAN

2.1 Needfinding activity with users

In terms of obtaining detailed, qualitative feedback, I seek to reach out to friends-of-friends via the snowball recruitment method. This should provide a relevant but manageable number of users to undergo live think-aloud play sessions. This process is ideally easy to perform considering there is a publicly available build of the game online from which gameplay could be shared live via Discord or Zoom. The participant would enter a screen-sharing call with me, be navigated to the game via their internet browser, and begin play. Beforehand, I will encourage them to voice their thoughts and decisions as they play (Appendix, 15.1.1). Think-aloud sessions would help provide insight into users' moment-to-moment decision-making process, as well as what elements may be surprising or unclear to them. These are incredibly important for the purpose of conveying helpful and relevant information via the HUD. After the play session, an interview with a handful of scripted questions would help guide their feedback, and prod at their opinions of the interface's usefulness (Appendix, 15.1.2). Ideally, this will provide a solid baseline of what I should be focusing on in terms of actionable improvements to the interface.

2.2 Heuristic evaluation

The interface that I will be evaluating is the HUD of the game (Appendix, 15.1.4). The HUD is the main interface through which users obtain moment-to-moment information about the state of the game and is visible at all times.

I plan to utilize the following heuristics to evaluate the interface: Discoverability, Mapping, and Perceptibility. All three are part of Nielsen’s 10 Design Heuristics.

I hope to follow the general structure of heuristic evaluation described in the provided material, “How to Conduct a Heuristic Evaluation” by Kate Moran and Kelley Gordon, and develop a similar procedure, detailed here:

- Ensure I understand the selected heuristics, preparing myself to examine the interface with these elements in mind.
- Play the game, making sure I am familiar with its functionality.
- Play the game, looking for issues (Appendix 15.1.5).
- Follow the Heuristic Evaluation workbook structure for each of the three heuristics, detailing the issues found and the relevant recommendations to be made.
- Compile information and account for possible exceptions.

3 NEEDFINDING RESULTS

3.1 Needfinding activity with potential users results

3.1.1 Think-aloud protocol results

Full transcripts are available in the appendix (15.2).

Table 1—Think-aloud protocol results, organized by issue and proposed recommendations.

Issues	Recommendations
Players didn’t know what was damaging them, from where, and took longer than expected to realize what was happening. Players are unaware	Directional damage notifications, on-screen display of current health and its changes need to be more visible. Make enemies more visually identifiable, and allow them to emit

of the presence of nearby enemies.	audio cues in order to alert players to their presence.
Player attempts to shoot while not realizing they still needed to reload. The player ends up reloading then they expect to be firing. Some players may expect an automatic reload to be performed after the clip is exhausted.	Notification from the interface to inform of need to reload, more visible ammo count, improved controls (adding the "R" key to reload). Consider the possibility of performing an automatic reload upon ammo reaching zero.
Player is confused after receiving a new weapon, unsure as to what they received. They are not certain as to how they can use/equip it. In several cases, players are unaware that they have obtained a weapon at all.	Better labeling as "new weapon: [name]" and instruct them to press the corresponding number key to equip, show mappings to number keys on HUD. Visualizing the arsenal of available weapons on HUD or suggesting the ability to switch may improve understanding.
Players are not being given adequate information about upgrades, leading to limited understanding of gameplay mechanics and little variation of upgrades. Player unexpectedly selected an upgrade by accident due to the interface.	Better labeling of upgrades to show their purpose and execution, as well as the effect of selecting the upgrade. Make it more difficult to accidentally select an upgrade with a confirmation button.
Players forget about the ability to sprint, or do not feel familiar with the bar representing remaining stamina.	Reposition the bar to a more conventional area on the screen. Provide more visual feedback than

	simply the scale of the bar, such as color and transparency.
Players do not understand the ability to charge up their shots.	Better labeling of the upgrade teaches this feature directly without tutorials. Stronger on-screen (UI) representation of the ability may improve understanding.
Players are having difficulty reading the HUD elements against the environment.	Improve contrast of HUD elements from the environment. Move HUD elements away from the extreme corners of the screen.
Damage varies across weapons in a manner that is not communicated or intuitive. Damage (across weapons) seems disconnected from the damage stat listed on the HUD. Weapon archetypes are not recognizable to players (i.e splitgun → shotgun). Starting weapon is considered the strongest, most reliable, and least annoying.	Relay the behavior and basic stats of each weapon, informing the user's mental model and allowing them to make informed decisions on how to best utilize their arsenal. Consider providing scaling information for each weapon (i.e. $DMG * 0.75 \times 3$ pellets for splitgun). Consider giving the obtained weapons greater unique strengths to encourage variation.
Users not informed of changes to enemy speed or behavior, growing difficulty.	Consider informing user of changes to behavior (i.e. "enemy speed up!" or "walkers emerging") to avoid undue surprises. Consider differentiated enemy designs corresponding to behavior.

3.1.2 Post-event protocol results

As of now, I have conducted four needfinding sessions with participants. As I perform more, I will make additions and edits to this section.

The participant expressed that at the beginning of the play session, they were “confused [about] how to do stuff,” but that they were soon able to get “into the groove.” Hopefully, some of the proposed changes will help teach the player how to succeed better than before. Players expressed that their performance was consistently “hampered by the lack of understanding of some of the mechanics.” A general feeling of “confusion” was shared by participants, who felt like they weren’t “well instructed” and that the general progression of the game was a mystery. Players generally felt that they didn’t know what to expect throughout the course of the game, with little time to explore or familiarize themselves with the mechanics between hordes.

Players also expressed that they “would have liked to get a little more” information provided to them, and that by being aware of certain mechanics, they would expect their performance to improve. One player noted that with practice they would get “better at selecting upgrades that are actually beneficial,” and that they didn’t know what they were doing.

A consistent source of uncertainty was the upgrade system. In particular, the “charge” and “luck” upgrades were a mystery to participants, who didn’t know what to expect upon obtaining them. While the upgrades seemed “genuinely worth grabbing and interesting,” their effects and worth were evaluated essentially at random. Participants expressed that they sought clarification on “what the upgrades might have done.”

They felt that the enemy salami “kills you a little fast.” The participant didn’t realize they were dying when an enemy would sneak up from behind, dealing damage to them. They provided a few suggestions to fixing this issue, such as making health more visible and intuitively displayed, as well as a visual/auditory

effect when getting hit. "I couldn't tell I was dying," said one participant, suggesting the latter of the aforementioned additions. Conversely, some players expressed the desire to be provided information on the health of the enemies themselves, whether in visible changes to the enemies appearances, or through 'damage numbers' appearing upon each shot connecting. Participants sought enemy behavior to be more easily expected, some suggesting "different colors" to denote variety.

Participants often discussed the difficulty of the UI "fading into the background a lot," making it difficult for them to pick out vital information. After their play session, a participant wasn't sure what the highest round they got to was, meaning that this HUD element may not have been easily readable. Another participant suggested stronger contrast between the HUD and environment.

The participant didn't know how to change weapons, trying the scroll wheel to no avail. They suggested making it easier to notice how to change weapons by teaching while unlocking (i.e. press 2 to equip), and would like to use the scroll wheel to change weapons in the future. In general, players weren't sure "what shortcuts were available" to them for the purposes of switching weapons. Similar concerns were brought up in the context of reloading weapons. In many cases, participants noted they wanted to see more weapon variety. I hypothesize that while weapon variety was readily available, the mixture of unlockable weapons underperforming and being unaware as how to be equipped contributed to this sentiment. One player expressed interest in "experimenting with using different guns for different strategies," but participants generally stuck with the most effective tactic available to them, opting to use the starting pistol. Participants seeking "more context on gun selection" suggests that this is partially an issue with the interface.

Another source of confusion lay in the behavior of the weapons and the participants' understanding of them. "I didn't understand that the 'Split Gun' was a shotgun," expressed one participant, expressing that they wished its weapon 'archetype' was clarified as a shotgun.

A participant noted that they routinely forgot about the presence of the stamina bar, suggesting it be moved “towards where the health and other information is” due to forgetting its placement near the top of the screen.

3.2 Heuristic evaluation results

3.2.1 Discoverability

Discoverability, or as it is referred to in Nielsen’s Heuristic Evaluation Notebook as “Recognition Rather Than Recall,” is centered around the theme of making relevant information readily available and visible to the user.

Table 2—Discoverability issues arising from heuristic evaluation and the respective recommendations of improvement.

Issues	Recommendations
• No controls are listed. • HUD elements are difficult to see against the environment. ◦ Many important elements are listed, but are so hard to see that it is difficult to refer back to them as needed. • “Equip [New Weapon]” is prompted, but the user is not informed as to how. ◦ Weapon mappings are not visible, must be remembered, and cannot be scrolled through. • Current upgrades are not explicitly visible, only through numerical stat changes, most of which are not visible.	• Provide basic control information at the start of the game. • Utilize the discoverability potential of the upgrades. • Ensure HUD elements contrast from the background for visibility. • When being given a new weapon, specify the number key (i.e. “Press 2 to Equip”) the weapon corresponds to. • Show which weapons are available and which are selected. ◦ Allow weapons to be scrolled through. • Make the list of current upgrades and their level visible to the player during gameplay.

3.2.2 Mapping

Mapping, or as it is referred to in Nielsen's Heuristic Evaluation Notebook as "Match Between System and the Real World," is centered around the theme of conveying information in a familiar, understandable way.

Table 3—Mapping issues arising from heuristic evaluation and the respective recommendations of improvement.

Issues	Recommendations
• "Glizzy Count" and "Damage" labels on the HUD are not specific or natural enough language to the user. • Upgrade names are inaccurate or confusing, their effects often being unclear. • Text is the only method used to convey information. ◦ Holding down the left mouse button causes "Bullet Charge: [% Full]" to appear.	• Use more familiar terms, or contextualize their usage better. • Provide short descriptions of each upgrade, or change their names. ◦ Unique icons would form an effective mapping between upgrade and effect. • Use different methods to convey information, such as iconography or meters instead of text.

3.2.3 Perceptibility

Perceptibility, or as it is referred to in Nielsen's Heuristic Evaluation Notebook as "Visibility of System Status," is centered around the theme of conveying the state of the system and providing feedback to the user.

Table 4—Perceptibility issues arising from heuristic evaluation and the respective recommendations of improvement.

Issues	Recommendations
• State of needing to reload is not made clear, the player expects to be able to shoot. • State of taking damage is not clear, the user receiving very little feedback. • Current strength level of the player is not made clear. • Destroying an enemy is given little emphasis or effect.	• An indicator or change of control options may prompt the correct reloading behavior. • Include directional indication upon taking damage, alerting the player to its source. • Visualize upgrades or strength rather than just numerically. • Provide more on-hit or on-kill feedback than a hit marker.

4 BRAINSTORMING PLAN

There are many systems to be improved, but the overall goal of my designs are to improve the communication of information to the player. For example, the needfinding exercises revealed that a major point of concern is that vital elements are not being made obvious or drawing attention to themselves. This will serve as a core goal that will support the overall goal of effectively communicating information to the player.

One of my major concerns for brainstorming is the possibility of conforming to certain standards or concepts that I have already been exposed to. I want to go into the exercise as open-minded as possible, while also reminding myself of many of the core issues unearthed in the needfinding results. Prior to the exercise

beginning, I will ensure that I am familiar with the most prevalent issues and seek to refocus my concepts around the overall goal by writing it down.

I will then engage in several roughly 10-minute bursts, writing down by hand any concepts that come to mind. The purpose of these short bursts is to help refresh and reset in order to prevent points from sticking around only one concept or area. However, these sessions will not have a hard limit, allowing me to continue the flow of ideas if needed. I will attempt to cover a broad range of subjects in these sessions, but if any element has gone particularly underrepresented in terms of concepts, I may refocus the next session starting with that topic.

I am a bit concerned about the possibility of confirmation bias interfering with the exercise, as I have several ideas and images in mind of what I had hoped things might look like, and so I am taking heavy precautions to mitigate its influence. I may engage in a burst centered around reverse brainstorming, a technique designed to help illuminate underrepresented sources of issues meant to be reversed into strong solutions (3). I believe this could help provide equal representation of previously undiscussed concepts. Additionally, a session with the support of ChatGPT may uncover new topics or areas to explore.

5 BRAINSTORMING RESULTS

I was able to give myself a number of dedicated brainstorming sessions as I planned, although I felt the effects of fatigue set in towards the end of the final session. I was trying to write down ideas that weren't identical to others in the past sessions, which contributed substantially to my cognitive load. Altogether, I feel the concepts succeed at providing ways for information and state to be effectively communicated to the user.

Certain topics centered around the concept of hierarchy. While brainstorming, I searched for the most important elements that need to be available to the player. These appeared in concepts such as increasing the size and contrast of elements on the HUD. This also involved a balance between which elements required the most amount of space, and therefore, attention. This culminated in a design

alternative that focuses primarily on the aforementioned readability and hierarchy. My mission was to retain much of the behavior of the previous design, but organize the information and present it in a stronger fashion.

I also wanted a manner that expressed the state diegetically. This would come in the form of physical signs of damage to the player, as well as the HUD elements existing in-universe. I wanted to group together several of the brainstormed features such as a flashing red effect for taking damage, an on-weapon ammo signifier, and a charging effect appearing on the weapon itself. This would communicate information in an in-universe way in order to make the gameplay itself feel more natural and intuitive.

The final design alternative was one that expressed many of the aspects in a themed manner. This design alternative would explore communicating info in novel, eye-catching ways. Most commonly, each informative element would be communicated in a way that afforded its behavior in some way. This would utilize certain brainstormed aspects including individual bullet UI, presenting the round number from an illuminated source, and utilizing a moving syringe to represent health.

Detailed results may be found in the appendix (15.5).

6 INITIAL PROTOTYPING

Firstly, I wanted to provide the current UI as a frame of reference for my suggested improvements via prototyping.

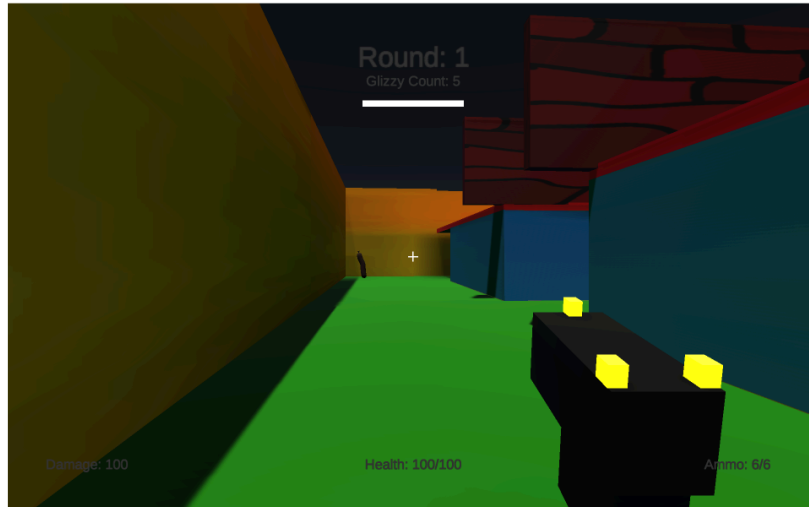


Figure 1—Base HUD in the original game *Salami ate my Neighbors!*

I first created the prototype for the bold, readability-based design alternative. This mostly involved maintaining the structure of the on-screen elements while also maximizing the amount of contrast and informational hierarchy I could provide to the HUD in general. I felt that this prototype was a massive improvement in terms of ascertaining the correct information from the screen.



Figure 2—The 'contrast' HUD, representing the need for greater visibility of elements..

I then created the prototype for the intuitive, diegetic design alternative. This involved inserting elements in-universe to make their function and meaning instantly understandable to the player. I wanted to include a way to monitor health and stamina with elements that made sense to the user, such as through a heart rate monitor. The bullets appearing on the side of the clip similarly convey the amount of ammunition remaining before needing to reload.

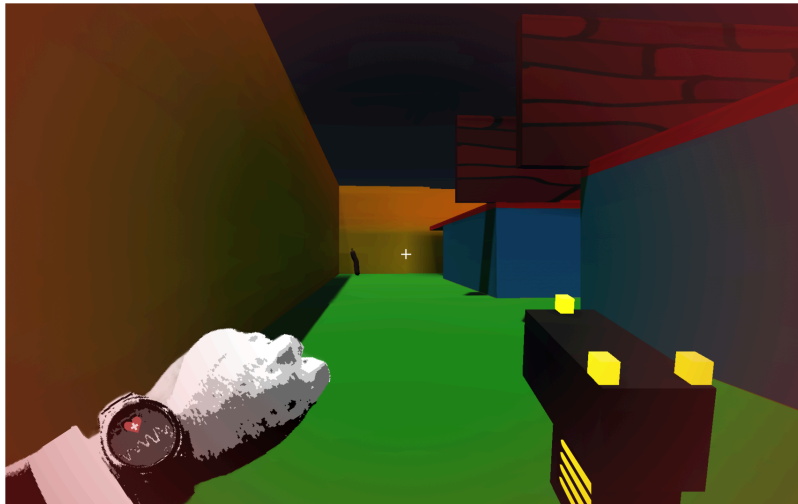


Figure 3— The ‘diegetic’ HUD, designed in efforts to leverage recognizable and intuitive behaviors.

The final prototype I developed attempted to use themed ways of communicating information to the player. This can be seen through the visual changes to the HUD, representing each particular piece of information in a unique way and through noticeable effects upon changing state.

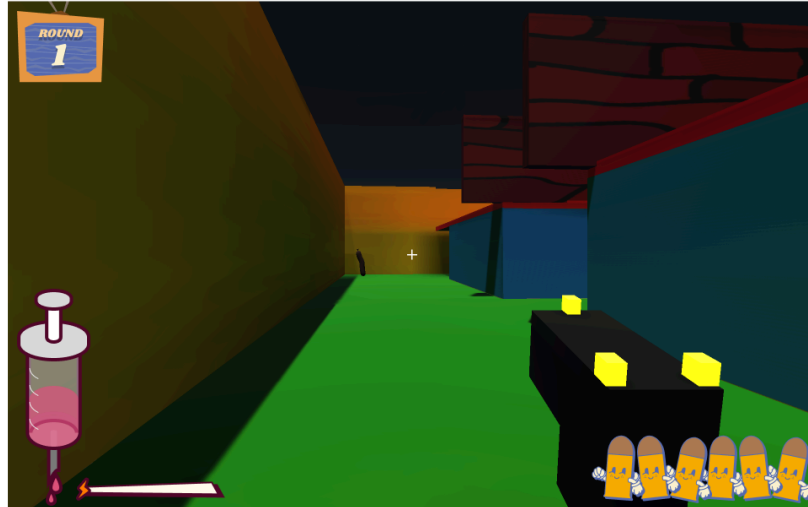


Figure 4— The 'thematic' hud, designed to use a number of styles which represent vital information differently.

7 EVALUATION PLANNING

I am creating a survey in order to determine the strength of the prototypes I have created as a result of the first prototyping stage.

The participants will be recruited through snowball and convenience sampling measures, with the survey being distributed through the PeerSurvey platform. This will make my survey very accessible to others and will provide me with the support I need! The participants will be asked to view the 4 HUD designs, compared against each other in pairs of two. At first, each of the proposed HUD prototypes will be set against the original UI to ascertain improvements. Then, the prototypes will be set against each other, in order to ascertain the strengths of each against each other. This will potentially reveal the strongest aspects from each design, informing the final prototype.

The survey is designed to derive qualitative and quantitative elements from the users' responses. Firstly, the users are asked which of the two interfaces they are presented with is stronger. Additionally, they are asked to describe the reasoning behind their decision, if possible. This means that the amount of selections as the stronger interface may be translated into quantitative data very easily.

Additionally, the reasoning behind the decisions will help add qualitative data to the results. These will back up the quantitative results, explaining what participants did or did not like about the designs.

The qualitative results will initially be coded to help combine similar comments and themes. From there, they will be organized by prototype design in order to evaluate which comments align with each design. Because qualitative responses were optional, we cannot develop a sentiment analysis from these comments alone. Instead, we will rely on the quantitative results to inform of overall sentiment, supported by the qualitative comments. The results will be summarized roughly by the themes identified in the coding process.

The quantitative results will be determined based on the responses of participants as to which of the two prototypes in each question that they believe is stronger. The results will be added up into the total selections of each prototype, from which statistical tests can be performed. The initial tally will give a rough idea as to which of the designs are favored, but by analyzing the results with a Bradley-Terry model, I can better determine the ways in which the prototypes fare against each other directly.

8 EVALUATION RESULTS

For the qualitative results, I was able to group the additional comments or clarification by common theme via the process of coding responses. A numerical representation of the feedback is visible in the following table. Positive scores are given to positive feedback on a subject, where negative scores are given to constructive feedback.

Table 5—Results of evaluation survey, presented as total sum.

Feedback Subject	Original	Diegetic	Contrast	Thematic
Information	4	2	8	6
Simplicity	2	1	2	0

Ease of Understanding	3	2	6	3
Usefulness	-1	-2	1	-1
Readability	-10	1	16	3
Obtrusiveness	4	-1	-3	-1

These results suggest several potential improvements that can be made in order to produce a quality final prototype. For instance, the ‘contrast’ prototype is recognized for its strong readability, perhaps at the cost of its obtrusiveness. The ‘original’ prototype is similarly praised for its lack of obtrusiveness, but its readability suffers. A common reason for selecting the ‘contrast’ prototype over others was that it was easy to read the text against dark backgrounds, something that respondents said the ‘original’ struggled at. If the qualities of these two prototypes can become combined to find a strong intermediate result, both can be maximized. When selecting elements present in the previous prototypes to be used, the corresponding qualitative feedback will be considered as to use the constructive criticism wisely and leverage the strengths of each design.

For the quantitative results, I first made a simple sum table of each of the results. Additionally, I constructed a win table for use in the Bradley-Terry model.

Table 6—Total values and win table constructed for use in Bradley-Terry model.

	Totals		Original	Diegetic	Contrast	Thematic
Original	15	Original	0	8	3	4
Diegetic	22	Diegetic	13	0	4	5
Contrast	46	Contrast	18	17	0	10
Thematic	43	Thematic	17	16	11	0

This provides us with the resulting ability scores, listed in the table below. These scores also communicate a somewhat conclusive ranking of strength. The corresponding ability scores represent each prototype's ability to win in a comparison against another.

Table 7— Ability scores of HUDs.

	Ability Score
Original	0.076108
Diegetic	0.115518
Contrast	0.416121
Thematic	0.392253

The Effect of Prototype on Ability Score

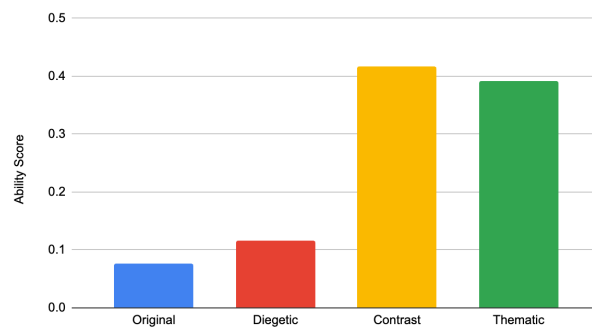


Figure 5— Ability scores of each prototype.

From the results of the quantitative analysis, we can see that there is a clear disparity when comparing the prototypes. The strength of the 'contrast' and 'thematic' prototypes are much greater and were chosen over the competition much more than the 'diegetic' or 'original' prototypes. These results suggest the ranking of Contrast > Thematic > Diegetic > Original, with 'original' being the weakest. This ranking itself does not suggest that all elements of the contrast prototype must be included going forward—rather, providing a helpful metric to gauge the overall strength of the combined elements of each prototype. In combination with the information gathered from the qualitative evaluation, these results will help inform the construction of the final prototype. By carefully selecting the strongest elements from the strongest prototypes, all the while avoiding the qualities that make the “weaker” prototypes fare worse than others, we can expect an overall improvement in terms of feedback from users.

9 SECOND ITERATION NEEDFINDING

9.1 Re-introduction

Receiving feedback on the prototypes I presented to survey participants was extremely helpful to the process of understanding users' needs. Participants' opinions on the prototypes began quickly revealing the strengths and weaknesses of each. The prototypes with the highest ability scores from the Bradley-Terry model also share the highest Information and Readability scores in the coded results. Despite having objectively the same amount of information as 'contrast,' the ability score of the 'original' prototype reflects its poor readability. What good is information if you can't read it? Similarly, participants seem to agree that the 'original' tends to obstruct gameplay the least— of course it does! It can hardly be seen in the first place. Nevertheless, the evaluation has revealed that even the strongest prototypes will have some inherent weaknesses— allowing them to be addressed and improved upon in future iterations.

There are no major needfinding questions that need to be addressed before moving forward to the next stage of the process. Instead, much of the effort will be spent using the results already gathered throughout the first iteration to create a stronger overall prototype, combining the strongest elements while eliminating those that are detrimental to users' understanding of state during gameplay.

Something that I was pleasantly surprised by from the first iteration through the design life cycle was that the amount of information displayed by the first HUD was generally seen as serviceable. The needfinding playtests relayed the message that users were generally lost due to the way that information was presented. In attempts to understand the best way to convey that information to the user, the prototypes were constructed. From there, I have been able to ascertain how to move forward.

There are absolutely some needs that I did not include in my initial prototypes, although I believe all of the major user needs have been accounted for. The purpose of the divergent prototypes were to cover lots of ground in an

exploratory way. Taking those results forward, we can address the users' needs more accurately.

The numbers and comments from survey participants alike show that there is not a single, decisive winner. Every prototype had some extreme results from feedback- positive or constructive- both being equally useful in determining the next steps. 'Contrast' and 'thematic' had the highest ability scores and were preferred by participants much more often, but that wasn't the case for all users, and certain aspects of those designs were considered much weaker than the same aspect of others. For this reason, the final HUD will combine elements from all prototypes.

10 FINAL PROTOTYPE

The following image represents the current version of the final prototype in its base state. One of the major needs of the interface was displaying the correct information to the user effectively in order to convey state. We'll first be looking at the base state because its elements are always on screen during regular gameplay. In this instance, we can see a number of pieces of information. Firstly, towards the upper left, we see a TV element, displaying the current round on its screen. Just below, there is a light orange line that we will later see representing the progress of the current round. Towards the bottom left of the screen, we see a few more important elements. In bold numbers, the current health is displayed. Additionally, a meter displaying the current stamina is also present just below the health. On the right side of the screen, we have a few more instances of information. The large numbers here denote the current and maximum ammo, with the little mascot and text describing the currently equipped weapon.

With all of this information available to the player at any given time, they will have the information they need to make informed decisions during gameplay.



Figure 6—The base HUD of the final prototype.

The next image is a bit more complex, showcasing several conditional elements that may or may not be present given the user's current situation during gameplay. It is unlikely that all of these elements would coincide at once, but all will be displayed for the purpose of discussing the elements.



Figure 7—The final prototype HUD with several conditional elements active at once..

Starting from the top once more, we see that the user is part way through the first round. In this situation, we see that the user is taking damage from behind and from the side, denoted by the red streaks pointing off of the screen. This is also reflected in their health being lowered from 150 to 100. At the center of the

screen, we see that the user has fully charged up a shot, being reflected in the red circle (which fills as the mouse button is held) being marked with a green stem, denoting its completion. Just below, we receive notice to reload– looking at the ammo count, we see that we are indeed running low! We can also see that the user is recovering stamina, from the darker color and shortened length of the stamina bar. Finally, at the bottom-center of the screen, we can see the number of upgrades that the player has collected with the purple one at the middle being at its second level.

This prototype leverages some of the strongest aspects of each design. For example, it maintains the bold numbers of the ‘contrast’ design. Additionally, it includes a red damage overlay that occurs when getting hit from the ‘diegetic’ prototype, albeit accompanied by a directional indicator that points the user to the source. To denote the weapon type, a little mascot is displayed within the ammo counter, straight from the ‘thematic’ prototype, and to describe the round number and progress, a TV icon is used. Finally, the general structure is still mirrored somewhat in the original prototype, which helped create the others.

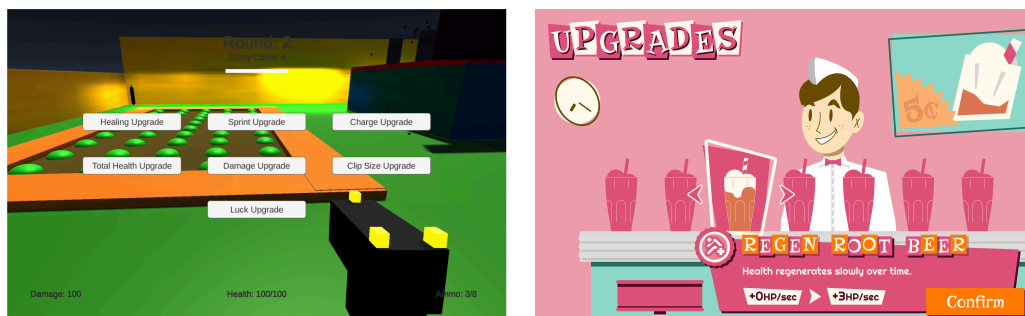


Figure 8—The before-and-after of the upgrade selection screen.

Additionally, I constructed an additional prototype to mirror the proposed changes to the upgrade screen. I include this side-by-side comparison to highlight the way in which information about the upgrades are presented. Instead of showing all the upgrades at once in minimal detail– a complaint of many users– I instead focused on displaying one upgrade option at a time in a horizontal list with all the details that the user may need. This keeps the user informed enough to make calculated decisions based on their needs.

11 VIDEO PROTOTYPE

https://youtu.be/HpmCj_CjHtY

<https://docs.google.com/presentation/d/1khdYpTMshg-VJK60SqJ22gavjASlB9zWds98JbqxxIg/edit?usp=sharing>

12 FINAL EVALUATION PLANNING

The final evaluation will be conducted among individuals via the snowball method, primarily those who have already been exposed to the project through the previous sections of needfinding and prototype evaluation. The participants will be asked to join a roughly 10-minute call with the incentive of their prototypes being evaluated in turn. The participants view the prototype's several "states" as will be described, giving their qualitative opinion and providing quantitative feedback on the strength of the HUD's representation of that state. The conversation will be recorded and the transcripts can be found in the appendix (15.6).

A slideshow will be constructed with the transitions of several notable state changes taking place, visible through respective changes to the UI of the HUD. These will include, starting with the base state of the final prototype: using ammo, taking damage, increase in round progress, round number progress, collecting upgrades, charging up shots, and switching the active weapon. The participant will be asked to anticipate what changes will be present in the UI when exposed to the "before" stage of the event. This will be a consistent source of specific qualitative data during the evaluation, which will be coded and organized by subject. The participant will then be shown the "after" stage, clarifying the actual result of the state change as reflected in the UI. After each interaction, the participant will be asked: On a scale of 1 to 5, with 1 being not very representative, and 5 being very representative, where would you place this behavior in terms of representing the behavior of [ex: taking damage]? Quantitative statistical analysis will be performed on the data, including mean, standard deviation, and interquartile range.

At this point, the participant will be shown another portion of the prototype, this being the upgrade selection screen. For this section of the final prototype, I constructed a slideshow with each of the abilities' information on display, navigable with the left and right arrow keys. This behavior would directly mirror the interaction of selecting an upgrade in-game. The act of using the interface will be shown to the users, and they will be asked: What would you consider the function of this screen to be? What are the obvious ways of interacting with this prototype? For users unfamiliar with the initial game/gameplay/HUD/UI, this may be a strong opportunity to show before-and-after transitions. Users may be prompted to provide input on how they feel about the change. These are similarly effective sources of qualitative feedback.

13 FINAL EVALUATION RESULTS

I was able to conduct interviews with five participants, recruited through direct outreach. I informed them about what would be expected in terms of commitment and they agreed, asking for no compensation. Two of the participants were engaged with the initial needfinding session.

I conducted the previously discussed statistical analysis on the data. This essentially relates to how well the interface is able to convey certain events or changes of state to the user. Results are viewable in the appendix (15.6.6).

The quantitative results and statistical analysis reveals several notable elements. In general, participants agreed that the interface was very representative of the state changes of firing a weapon, taking damage, and advancing rounds. In contrast, the averages suggest that the interface's ability to represent eliminating an enemy or obtaining an upgrade to be roughly neutral- neither particularly representative nor unrepresentative. The standard deviations and IQRs suggest that obtaining upgrades was the least agreed-upon in terms of representation strength, followed by eliminating enemies, and then charging weapons. This communicates that while the interface's ability to represent these elements was not the strongest overall, it was not exactly agreed upon by participants. For example, one participant rated Obtaining Upgrade, as '2 - Very Representative,'

while another rated it '-2 - Very Unrepresentative.' This disparity is not detectable by studying the mean alone.

The qualitative results were coded and organized, with results being found in the appendices (15.6.7). The results show that very similar expectations were held by all participants for the behavior of firing weapons, taking damage, and advancing rounds. Additionally, a secondary behavior or change relating to these events was also anticipated by more than one participant for both taking damage and advancing rounds. This shows that the behavior of these two changes to the interface are easily anticipated, even by a new user. For example, all five participants expected a change to the number in the bottom left of the screen when taking damage. Two participants also expected to see a flash of red on-screen as well. Both of these changes are present in the actual HUD. It is worth noting that the least variability of coded responses was present in the firing of weapons, where all participants were able to correctly identify the sole change of the amount of ammo remaining. In contrast, charging weapons had the most different subjects of response, in which the most common element of a decrease in ammo count was mentioned by only three participants. This shows that participants had a difficult time finding an obvious, common explanation for the described change in behavior. The reveal of the actual change in the HUD was notably surprising for all of the participants, some of which mentioned that they would prefer it handled differently.

14 REFERENCES

1. Caroux, L., & Isbister, K. (2016). Influence of head-up displays' characteristics on user experience in video games. *International Journal of Human-Computer Studies*, 87, 65–79. [doi:10.1016/j.ijhcs.2015.11.001](https://doi.org/10.1016/j.ijhcs.2015.11.001)
2. Caroux, L., Le Bigot, L., & Vibert, N. (2011). Maximizing Players' Anticipation by Applying the Proximity-Compatibility Principle to the Design of Video Games. *Human Factors*, 53(2), 103-117. <https://doi.org/10.1177/0018720811400600>
3. Caroux, L., Delmas, M., Cahuzac, M., Ader, M., Gazagne, B., et al.. Head-up displays in action video games: the effects of physical and semantic characteristics on player performance and experience. *Behaviour and Information Technology*, 2023, 42 (10), pp.1466-1486. (10.1080/0144929X.2022.2081609). <[hal-03681361](https://hal.archives-ouvertes.fr/hal-03681361)>
4. Kitch, B. (2024, August 15). How to use reverse brainstorming with your team: Mural. Mural. <https://www.mural.co/blog/reverse-brainstorming#:~:text=Reverse%20brainstorming%20is%20a%20problem,that%20tendency%20to%20your%20advantage>
5. Moran, K., & Gordon, K. (2024, August 2). Heuristic Evaluation Workbook. Nielsen Norman Group. https://media.nngroup.com/media/articles/attachments/Heuristic_Evaluation_Workbook_1_Fillable.pdf
6. Moran, K., & Gordon, K. (2024, August 2). How to Conduct a Heuristic Evaluation. Nielsen Norman Group. <https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/>
7. Peacocke, M., Teather, R. J., Carette, J., MacKenzie, I. S., & McArthur, V. (2018). An empirical comparison of first-person shooter information

displays: HUDs, diegetic displays, and spatial representations. Entertainment Computing, 26, 41–58. [doi:10.1016/j.entcom.2018.01.003](https://doi.org/10.1016/j.entcom.2018.01.003)

15 APPENDIX

15.1 Needfinding activity preparation

The primary needfinding activity is structured as a two-part qualitative session. Prior to the session beginning, participants will be informed that their play session and the subsequent interview will be recorded and that notes will be taken. The participants will be informed that their identities will remain anonymous for the purposes of the study.

15.1.1 *Think-aloud protocol*

Just prior to the think-aloud protocol beginning, I will let the participants know that I am hoping to record their thoughts during play, and encourage them to vocalize their reactions.

I will inform the participants of the intended controls and interaction method. This entails using a keyboard and mouse setup: WASD to move, the mouse to look around, left mouse button to shoot, and shift to sprint. I will ask participants if they have used this control scheme before, in order to ascertain their level of experience with the basic control method.

I will inform users that their goal is to survive the horde of salami.

After at least three games' worth of play (roughly 5-10 minutes depending on skill) and upon losing, participants will then be moved on to the post-event protocol. If during a game, the participant encounters a game-breaking bug, the game will be restarted.

After the study is complete, a transcript of dialogue and corresponding actions/events in game occurring during the think-aloud protocol will be constructed.

15.1.2 Post-event protocol

The post-event protocol will be conducted immediately after the think-aloud playtest. The following questions will be asked to respondents:

- How did you feel during the game?
- How do you feel about your performance?
 - How do you think practice might influence your progress?
- Does this game remind you of any other games?
 - If so, which ones and why?
- What is your perceived tone of the game?
- How did you feel about the amount of information provided during the game?
- Did you find any parts of the game unclear, or did everything make sense to you?
- If you could change anything about the game, what would you change?
- If you could add anything to the game, what would you add?
- Did you have any thoughts or feedback about the game?

Additional Questions

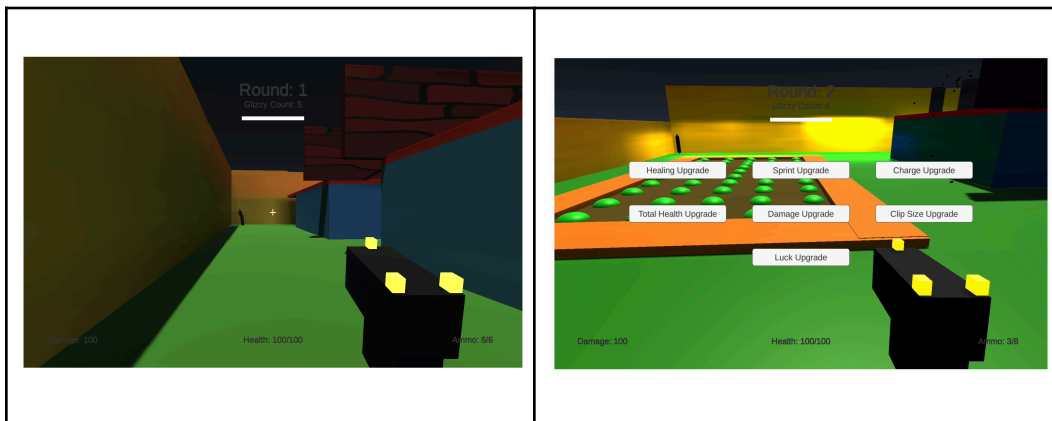
- How would you describe the game to someone with similar game taste to you?
- How would you describe the game to someone who doesn't play video games?

This is tricky as I am aiming to completely avoid leading and closed questions, as well as not make it too obvious that I am hoping to obtain feedback related to the interface.

15.1.3 Secondary activity

Should additional feedback be required, a survey may need to be constructed in order to obtain a large volume of feedback in hopes of gauging public opinion on certain aspects of the interface. The main participants would be people on Reddit, likely members of the r/CODZombies and r/BTD6 subreddits, with the Zombies and Bloons communities being target audiences for the game. Both communities are familiar with round-based survival games, intensely dedicated, and would likely enjoy the engaging gameplay loop of *Salami ate my Neighbors!* However, the ability to reach out to these audiences is not certain due to posting restrictions on the respective subreddits. Therefore, surveys may need to be distributed by other methods, likely to classmates or through snowball recruitment of friends in the gaming community. This is also uncertain, but through these methods a ~20 person total should be able to be met. In all, utilizing surveys as a needfinding method proves to be non-ideal for this situation.

15.1.4 Heads-up display



15.1.5 Heuristic evaluation questions

- When I make an error, what could've prevented it?
- What information am I missing as the player?
- Is the information presented in the best way?

15.2 Needfinding activity execution

15.2.1 Needfinding activity session one

15.2.1.1 Think-aloud protocol– “Oh! I died.” The participant attempted to shoot at the small target of a salami in front of them, backing up as they attempted to connect their shots. However, as they stepped back, they likely ran into a salami behind them, killing them. They were not alerted to the presence of damage behind them, leading to their death.

“I’m about to die.” The participant recognizes that their health is low and adjusts gameplay accordingly, backing away from the approaching salami.

“Uh-oh. There we go.” The participant attempts to shoot a salami, not realizing they were out of ammo and needed to reload. The gun reloads when the participant expects a bullet to be fired.

“What does this do? Splitgun. I don’t know. Or did I read that correctly?” The participant is informed by the interface that they have unlocked an additional weapon. However, they are unsure what that means and how to use it, or if they are even using it in the first place.

15.2.1.2 Post-event protocol– *So now I just have a couple questions I’d appreciate if you could answer. So how did you feel during the game?*

When I was first starting to play, I was a little confused as to how to do stuff. I feel like the salami kind of kills you a little fast. But once I started actually, like, getting into the groove, it was pretty fun. I did tend to find that the extra weapons, if you didn't notice, I didn't find that the extra weapons, like the split shot, I feel like it was very hard to actually hit stuff with the split shot versus the

normal pistol. So I ended up not using it, even when I probably could have. It was just kind of hard to hit with the bullet size, and the auto rifle was good as well. But I feel like once I upgraded the clip size of the pistol, I felt like the auto rifle kind of didn't really have a point much until I could upgrade it again, which was fine, because I probably would have upgraded again for the next round, but that round, it kind of felt a little small. Overall, though. I did find it. I did find it very fun. I had to close a few of my tabs, I had like 14 tabs open, kind of my fault.

All right. The next question is, how do you feel about your performance?

Oh, me? I was the greatest person to ever play this game. Obviously, no, no, but I wasn't particularly good. I haven't played too many first person shooters, especially, I guess was technically a round shooter. I don't know what they're called, but yeah, but I haven't played too many of them. So I'm not particularly good, but I played a few of them before on Minecraft, so I know the gist, but I think it did okay. Got to round five, was it?

How do you think some more practice might influence your progress or skill?

I feel like once I, if I keep playing this game, I feel like it could actually very easily get to round nine and get out the door.

Does this game remind you of any other games?

Yeah, there's a few of the Minecraft mini games that kind of were like this, where. And by Minecraft minigames, I mean the minecraft game that's based on COD zombies, I haven't played COD zombies, so I only know the Minecraft version of it.

Why did it remind you of the Minecraft games?

Just the gameplay loop of: you defeat a wave of enemies, then you get some upgrades. It reminded me pretty much what the Minecraft one was.

What was your perceived tone of the game?

Very silly, because there's a bunch of moving sausages running at you. It felt like it was serious because, you know, you were, you were getting killed by them, but also felt very, very silly because they were literally sausages moving like little inchworms.

How did you feel about the amount of information provided during the game?

I feel like the information was good, though I didn't really know how to change my weapon until you told me, because I didn't know to press one, two and three to get the additional weapon. I did try at one point, you couldn't see it, but I tried to use the scroll wheel to try to scroll between the weapons, but it didn't work. I know in a lot of games like that, you can use the scroll wheel like in CounterStrike. You can use your scroll wheel to go between the knife, the side arm, and the main weapon.

Did you find any parts of the game unclear, or did everything else, or did everything make sense to you?

Pretty much everything made sense. There's a few times where, when I was dying, I didn't quite realize I was dying, because your health going down is pretty small in the middle of the screen. Like the health being a little bit bigger might help, because the clip size and all that that was, that was all fine, but the health I would occasionally not notice when a guy snuck up behind me and started beating the crap out of me, and I wouldn't notice.

If you could change anything about the game, what would you change?

I guess, the things I would change would be, I'd make it somewhat easier to notice how to change your weapon, whether that means, like, adding the scroll wheel functionality I talked about, or maybe just saying, like, press two or press three when the weapon pops up and you learn it. And then the other thing I might add is some sort of like, damage effect on the screen when you get hit, maybe you could even play a funny sound.

If you could add anything to the game, what would you add?

Ooh, add? One thing that I might want to add is, I don't know if this weapon would be possible, but there were some times when I'd want to take care of a large group, but I didn't really want to aim my shotgun. Something like a grenade might be really, really interesting.

Do you have any general thoughts or feedback about the game?

Other than what I was just saying, I don't think so.

15.2.2 Needfinding activity session two

15.2.2.1 Think-aloud protocol– The participant spends a long time scanning upgrades, moving their mouse over each one, before selecting “clip size” upgrade. This is one of the upgrades which is named in a self-explanatory manner. By moving their mouse over each one, they could be looking to reveal more information or context regarding the upgrades.

“I don't know, just buy the clip,” the participant said, selecting the “clip size” upgrade once more. This is purely a hypothesis, but I propose that the reason for upgrading clip size multiple times is due to reloading being so restricting and unnatural during gameplay.

After selecting a damage upgrade and speaking of the upgrade screen, the participant asked: “what is, I don't know– or, understand what, like, all of those things do, like, what is ‘luck?’ What do you even use that for?” The next time, they again asked for clarification: “yeah, like, I understand healing, total health, sprint, damage, clip size, but what do like, charge upgrade and luck upgrade do?”

Later, the participant accidentally selected an upgrade as the prompt pops up on screen very briefly, causing them to exclaim “oh, oops.”

After being given the “Equip AutoRifle” prompt, the participant cycles through weapons, saying “oh, what? I didn't even know you could switch weapons.”

“I forgot there is a shift,” says the player, recalling their ability to sprint.

Discussing the difficulties they are facing during gameplay, the participant relayed that “[the enemies] don't make any noises. So they just sneak up on me, I'm just better off just sitting in the corner.” Subsequently, an enemy spawns in the corner behind them, damaging the player unexpectedly. “Oh, never mind.”

Attempting to charge shots before firing, and unintentionally shooting a shot before it begins to charge, the player says “[the weapon] shoots, even though you want to charge [up a shot].”

15.2.2.2 Post-event protocol– *So first off, how did you feel during the game?*

I felt engaged. I wanted to keep playing.

How did you feel about your performance?

I think it was good. It was just hampered by the lack of understanding of some of the mechanics.

Do you think some more practice would influence your progress?

Yeah, now that I'm aware of some of the stuff, like the charge and stuff.

Does this game remind you of any other games?

Yes.

Which ones and why?

Call of Duty Zombies, for obvious reasons.

What are the most obvious tells about that?

It's a round base shooter against a horde of enemies,

What is the perceived tone of the game?

Lighthearted and funny.

What makes it lighthearted and funny, to you?

The enemies themselves, the writing stuff on the wall, the environment in general.

How did you feel about the amount of information provided to you during the game?

I would have liked to get a little more.

Did you find any parts of the game unclear, or did everything make sense to you?

The charge mechanic and the luck or like, the charge and luck upgrades, I didn't know what they did.

If you could change anything about the game, what would you change?

I guess, a weapon and enemy variety, although I didn't get very far.

If you could add anything new to the game, what would you add?

More areas to run around in. Also a damage indicator, or an audio cue to show where the enemies are. I couldn't tell I was dying.

And do you have any thoughts or general feedback about the game?

I don't think so.

15.2.3 Needfinding activity session three

15.2.3.1 Think-aloud protocol– “The darkness of the background, like the skybox, is kind of causing me to have a little bit of a problem seeing the ‘round 2, glizzy count’ thing.”

“Yeah, I didn't notice– I think I've seen it now. There's a stamina bar at the top.” This is mostly due to the quirk with the UI of their screen recording software.

“I think I have a little bit of an issue seeing some of the UI because of the shadows and whatnot.” The participant feels that the UI is hard to distinguish from the background and other elements on screen

"It's showing three different particle effects. So I'm not quite sure how it shoots—like, if it's shooting in a cone." The participant mentions the splitgun (shotgun) and its ambiguous behavior, confused as to how many bullets/particles are fired upon each click and in what pattern.

It does less damage than the other things? It says 'damage: 100' [on the HUD]." The participant expresses further confusion about the behavior of the splitgun (essentially a shotgun) and its low damage, against convention.

"These [enemies] are usually one-shot, [but] when I go to this (the splitgun) sometimes it'll just... [take more than one shot]. So it's like a shotgun? Is that what it is? [...] Yeah. I also didn't fully understand that this was a shotgun when I got it, but that could very well just be my bad."

15.2.3.2 Post-event protocol— How did you feel during the game?

I did enjoy going around and just surviving waves. I feel like that's always an enjoyable panic of just, like, wave survival. I remember a long time ago, like one of the first games I ever played as a kid was, like a wave survival game. I forget exactly what it was called but it was like a zombie game, the same sort of thing, and it was very fun. So I think I like the core game side loop, and I like the idea behind most of the upgrades. I went to the upgrade screen, it looked like there were a lot of options. And I, I feel like in a lot of games there will be options that I, I'm very quick to just like, I don't care about them at all. But a lot of those options seemed like they were genuinely worth grabbing and interesting to like, try to go down. The only things I kind of disliked were, you know, the— the movement errors, you know, UI kind of fading into the background a lot. Yeah, it was mostly just movement that was the problem for me. I felt like the level had, like, I liked, there was like, diverse layering, so I actually could— when going through the level, it felt like there wasn't a singular area that it was like, oh, I'll just stay here, corner myself and then you'll be fine, and there's no way that

anything could kill me, like there wasn't like a super easy place to just stay the entire time, it encouraged me to move around.

How did you feel about your performance during gameplay?

I felt okay about it. It felt like for the good amount of it that seemed a little bit out of my control, since it would just like, like, as you saw, it seemed to, like, buffer some of my inputs, and was having me shoot across the map and things like that. But overall, I felt okay about it. I probably could have done better if I was in my normal setup, that is my bad. I didn't fully understand that I was going to test it right now. Had I known that, I would have been at my desk, at my computer. I was enjoying it throughout it.

Does this remind you of any other games that you've played?

An old school Flash game, Boxhead. That's what it was called, it reminds me a lot of Boxhead, just like the gameplay loop of zombies coming after you and you having to have a bunch of upgrades and different guns that you can acquire. Definitely reminds me a lot of Boxhead.

What was your perceived tone of the game?

It seemed like a kind of a weird, dystopian, glizzy kind of environment? So probably like, maybe a little bit more satirical, I would guess.

How did you feel about the amount of information provided to you during the game?

I definitely felt like a good amount of it could have been a bit more clear, mainly, I think being able to understand exactly what the upgrades might have done, or, like, I didn't understand that, like the split gun was like a shotgun, like some of that stuff was misunderstood. So I think some things could definitely be clearer.

If you could change anything about the game, what would you change?

I think I would call the split gun just a shotgun. I would try to change the UI to make it probably contrast more with the environment around so that it's easier to see and probably move the stamina bar towards where the health and other

information is, because often I would forget that it was up top when I was looking at other stats.

If you could add anything extra to the game, what would you add?

I think it's just my bad. I think maybe I did not just– I didn't get far enough, but more guns, like a little bit more variety, and maybe getting to choose different weapons instead of– I think I only got to experience the split gun, so getting to choose a different option would be nice.

Did you have any thoughts or feedback in general about the game, aside from what you've already mentioned?

I think I mostly covered it all.

15.2.4 Needfinding activity session four

15.2.4.1 Think-aloud protocol– “...assuming I gotta kill that” shooting the enemies creeping towards them

“health is not going well” they lose health unexpectedly, early into the game and as a result says “All right, [I’ll] take the healing upgrade.”

“I’m guessing, if I feel like you see barrels in games, you can blow up and not that one, right?” The participant points out an inconsistency with convention or what some may consider ‘video game logic’.

“I got five left, 4,3,2,1, reload. It's kind of annoying that it doesn't auto reload.”

“Uhhh... luck. I don't know what that means. Let's go with that.” The participant picks the upgrade to luck but doesn’t understand what the outcome of their action will be.

“All right, let's get [the autorifle]. It keeps saying equip, but I don't know how to.”

“Oh, he's running [...] it didn't let me know that they're not going to be crawling anymore. So that's unfortunate.”

“Oh, dude, I still have glizzies on the board. I gotta take these glizzies out.”

“Ah, how do I do other things? Like it's just talking about it like a split gun. [...] So it's more like a shotgun or a rifle.”

“That's actually a lot. I'm just gonna switch back to the pistol. Just because [the splitgun is] so loud. Also, I like how the pistol one-shots.”

“I lost. Oh, there was one behind me.”

15.2.4.2 Post-event protocol– How did you feel during the game?

Initially curious, and then it was a little whimsical at first. But honestly, I feel like the progression of the rounds wasn't super well instructed. So I kind of just felt like, Oh, I could potentially be doing this for like, rounds and rounds, and I don't know it's kind of foreboding, just getting attacked by burnt sausages for a bit. So, yeah, I guess how did I feel? It was just kind of like, confused.

How did you feel about your performance?

Um, I don't know whether it was a screen share or what, but the latency was pretty bad. So I felt like I couldn't really like, move and shoot the way I would want to, like, you know, Doom style or TF2 style game. I felt like I did, okay, yeah, okay.

Do you think that practice might influence your progress or aptitude at the game?

Yeah, for sure. I would know, like mechanical practice. I would hit, just hit more. Probably would experiment with using different guns for different strategies, like figure out if the rifles may be better for that fifth round where I was getting attacked by the really fast ones, and then also just getting better at selecting upgrades that are actually beneficial, because I didn't really know what I was doing.

Does this game remind you of any other games?

Yeah, definitely. TF2, kind of Source-ish type games. I think it was like the pistol kind of reminded me of just like a stock gun sound that you would hear in a Source game.

What was your perceived tone of the game?

it was kind of a mishmash, because it's like, whimsical, but also a little bit, like off putting, because the colors are, like primary colors, like, dude, you got the green garden and then, like that, like, kind of ugly, brown, gray wall. So the tone is whimsical, but also I'm terrified for my life, and I'm being hunted actively, so a little bit of both.

How did you feel about the amount of information provided to you during the game?

I definitely wanted more like, maybe, like a countdown to like, some games will have, like, a transitional period between the rounds where you can say that you want to move to the next round and like, it'll let you move around the map. And that kind of gives you a chance to like practice or like practice movement, or explore the map before you continue. That would have been nice. And then also more context on gun selection and, like, what the next round would be, maybe? Yeah, because it just kind of give you the count, but it doesn't necessarily say that, like, oh, now they're going to be running at you.

Did you find any parts of the game unclear, or did everything kind of make sense to you?

I wasn't sure if there was any differences in the hitboxes for the glizzies. Like, whether headshot would be better, or is the same as a body shot, right? Also, maybe I just wasn't looking clearly enough, but I didn't notice the damage differential. Like, if it said, some games will show you, like how much damage was inflicted on each shot. Like TF2 has a mod where it'll show the numbers actually popping off of the hurtbox that's getting hit, which is helpful to figure out how much damage you're actually outputting. I don't think that this game had it, and if it did, then it was kind of hard to see.

If you could change anything about the game, what would you change about it?

Yeah, we talked about the latency. Definitely would, I don't know if that's, that's just a my-machine thing. Definitely wouldn't want the latency to be better. Maybe. Clear controls on how to like switch [weapons], because some, some games will have it where you can use Q and E to switch the weapons. I just wasn't sure what shortcuts were available for actually doing inventory changes I see, and then also maybe, like a reload button. I actually can try hitting R, but some games will reload with R. In this game, you can only reload when you were out of bullets, which isn't intuitive for me.

If you could add anything extra to the game. What would you add?

I mean, I probably just didn't get far enough to see different enemies, but different enemy sprites with either, like, different speed or maybe different colors. Yeah, just a little more variety in the enemies.

Do you have any other thoughts or feedback about the game?

Just using TF2s, like UI, there's some really cool user mods just to make it, like, a lot more, like, visually salient and like, about, like, what's going on with my like, my clip, and like, how much damage I'm outputting and then also my health like, there was times when I had to, like, completely look into the corner and read the health bar. There is no color indicator or any sort of, like, classic finding game health bar where it's, like, I can tell that it's low because it's not very full. I had to actually, like, read the text to figure out what was, what the status of my health was. I think that probably would be helpful too. You know, all things that I'm sure you've thought of already, but, yeah, those are my thoughts.

15.3 Literature review

Finding difficulty locating relevant papers in the basic ACM Digital Library Search, I began looking into CHI PLAY, the premier conference for games UX. This proved to be a strong resource for later, but I pursued a general initial paper to provide me with a broad sense of direction. *Influence of head-up displays'*

characteristics on user experience in video games by Loïc Caroux and Katherine Isbister serves this purpose well. Published in 2015, the paper notes how studies have rarely “investigated the influence of HUDs on player behavior, and more specifically on player experience.” This makes the paper an ideal candidate to help inform my growing understanding of strong interface. The study provides a useful summary of the current understanding of HUDs, noting that “even the most intrusive” HUDs are “accepted, even desired” by players. “The main condition for high acceptance is that displayed information should be useful for the player,” perhaps meaning that extraneous information should be explicitly avoided. Furthermore, I infer that this finding suggests that the more important elements should be given more space and presence.

This paper also exposed me to *Maximizing Players’ Anticipation by Applying the Proximity-Compatibility Principle to the Design of Video Games* by Loïc Caroux, Ludovic Le Bigot, and Nicolas Vibert. This paper discusses the “proximity-compatibility principle,” which states that “important contextual information should be placed close to the main anticipation area of the game.” However, this only remains true if the contextual information does not overlap and obstruct the anticipation area. Essentially, this reminds designers to place important information near where the players are looking, but not in front of it. *Influence of head-up displays* specifies that “pieces of information used in isolation can be placed farther apart.” Searches into SEEV, discussed by this paper, revealed the study *Head-up displays in action video games: the effects of physical and semantic characteristics on player performance and experience* by Caroux et al. which essentially finds that player performance is only significantly diminished when main elements of the HUD are removed.

In terms of results, *Influence of head-up displays* found that for the FPS game they studied, CoD 4: Modern Warfare, the most eye fixations on the HUD occurred on the compass element, positioned at the bottom center of the screen. This is where information such as perks are visible in the modern HUDs of CoD: Zombies. However, it should be noted that the compass likely provides much more valuable moment-to-moment information than the list of active perks. The

participants of the study noted that “ammunition and available guns were not important elements” by comparison, by both novices and experts. This may mean that designs may seek to de-emphasize these elements on the HUD. The study also notes that novice players tend to fixate on temporary banners or “messages” that flash on screen. This may suggest that players find these elements distracting and should be avoided unless capturing the player’s attention is indeed the main objective. The players noted that the HUD elements were too far from the “main action area,” which is visible from the game view.

The permanent HUD elements are locked to the bottom of the screen, essentially half a screen away from the targets that the players will be focusing their attention on. In the case of *Salami ate my Neighbors!* we may find that due to the lower stature of the targets, the elements of the HUD may need to stay closer to the bottom of the screen, while still remaining outside of the “anticipation area.” Another interesting finding of note is that novice players stated that they did not use the HUD during gameplay. This is surprising considering that “novice players actually used HUDs,” as much as expert players.

The study revealed two “critical characteristics of HUDs,” being composition and spatial organization. In the words of the study, “FPS HUDs could be optimized by displaying elements as close as possible to the main action area for FPS games.” In efforts to better understand the effects of HUD layout on playing, they developed a subsequent experiment based on the results of the first, hypothesizing that by placing the main element (compass) at the top, closer to the area of anticipation, players would be given a better experience. The results show that the full, original HUD was rated the highest by participants, but that the layout with the main element at the top of the screen was rated second. The paper acknowledges that “the explanation could be that participants preferred HUDs where all contextual information was displayed in a line at the bottom of the visual interface.” Overall, the study encourages a persistent HUD with elements aligned across the bottom of the screen, adhering to the proximity-compatibility principle otherwise.

Pursuing more specific details, I searched for papers citing the previous study. This brought me to *An empirical comparison of first-person shooter information displays: HUDs, diegetic displays, and spatial representations* by Peacocke et al. The abstract of the study offers several of the conclusions (listed below), but I wanted to see if there was any other information I could gather.

Findings from *An empirical comparison*:

- “No one display type – HUDs or alternatives – are universally best”
- Ammo: “player performance was best with diegetic/spatial displays”
- Health: “players performed significantly better with a HUD”
- Weapon displays: “results were best when showing a redundant HUD icon and a diegetic/spatial display (the actual weapon)”
- Navigation: “A spatial ‘navigation line’ (showing the path) was best, but HUD-based mini-maps offered competitive player performance”

Something worth noting for their findings regarding the health display was that the numeric-on-HUD had the strongest performance, but participants preferred the bar-on-HUD. This is what informed their conclusion of participants preferring a HUD for health. Perhaps combining the two, as many modern games have, is the strongest solution? The rest of the findings were communicated well via the abstract, and serve as valuable points of reference.

15.4 Expected improvements

Before an evaluation study is designed or initiated, I wanted to create some hypotheses regarding the potential participant feedback and actionable improvements that can be made to the current user interface.

One of the most glaring usability flaws is that HUD elements can be difficult to read or missed entirely. “Round 1,” “Glizzy Count: 5,” and “Damage: 100” are particularly challenging to pick out in this photo, and overall HUD readability

changes drastically during gameplay. This HUD is certainly functional when visible, but there are certain changes I expect will need to be made as well.

Another element, or lack thereof, that I expect will receive some comments is the damage indicator. While there is a text element displaying health, instances of taking damage are not made obvious to the player. This can lead to unexpected deaths because damage is coming from an unexpected source or direction. It is possible to succeed without this HUD element, but when excluded it limits the amount of useful feedback given to the user.

An aspect of gameplay that personally created a surprise for me was that the game has very little indication of when reloading is required. Because there is no dedicated key to reload (often relegated to "R"), reloading can only be performed by attempting to fire with an empty magazine. Because there is little feedback or supplemental information to convey this behavior to the player, I expect players will be looking for a change here. While this action can be worked around, an indicator or change of control options may provide the appropriate feedback or flexibility that can allow the user to feel more capable.

15.5 Brainstorming exercise transcript

This is the transcript of each burst from the brainstorming session.

Improve the Communication of Information to the Player

Burst 1:

Large bold lettering with outline to provide contrast

Visualize clip size with individual bullet UI

Large number for current ammo, small for max ammo number

Translucent HUD elements when salami nearby (*for better visibility of low targets*)

Blood dripping down from top of screen when taking damage

Reveal round number and info with light-up sign from diner or theater

Create excitement effect around ammo or health when large changes occur

Create dedicated screen and explanation upon acquiring a new weapon

Directional damage indicators at the top of the screen like a compass, as well as at the middle of the screen around the crosshair

Sprint/stamina meter sweating when low

Sprint meter transparent, changing colors when increasing/decreasing, solid and white when full

Show ammo count diegetically, through number (*on gun*) or physical bullets in gun, maybe even an icon or light bar that decreases with ammo

Charge shot → effect around cursor, creates a full circle when complete

Charge shot adds effect to weapon → light/charge at the tip of the barrel, lightning effects, or neon bars illuminating

Health being shown only through diegetic red screen

Health shown as bar

- Syringe depleting with health, spurting liquid as effect

Ammo count raining bullets (*shells?*) upon firing for emphasis

Health and ammo counters shown as flip boards or spinning counter tally things

Health/Ammo shown as signs that light up and go dark/spurt sparks when taking damage

Weapon name/type shown through icon

- Represented by cartoon character on screen that pops up or walks into frame upon selection
- Shown as row/col of little tabs that pop up when cycled to
- Shown as revolving circular selection
 - Either side or screen, or temporary menu that pops up quickly (*at the center*) for radial selection

Burst 2:

Sound cues

- Upon weapon being charged up fully (ready to release)
- Upon round change
- Music intensifies/drops [off] with the number of enemies on the map (no hud presence)
- Hurt SFX to alert player taking damage

Enemy meter to show progression through each round (similar to Plants vs Zombies)

Round # communicated in different ways

- Roman numerals → numbers (CoD Zombies)
- Label, but better readability
- Need to keep readable but not distracting
 - Does it need to be checked constantly? Probably not, so maybe not constantly visible?

Tell the player what type (or archetype) of weapon they are gaining, done through text? Image?

Show damage type (i.e. energy, projectile), behavior (i.e. semi-auto, auto, single-shot), and damage itself (i.e. 20/shot, 5 x 10/shot)

Press dedicated weapon slot number to equip (i.e. "Press 2 to Equip Splitgun")

Show damage increase visually/audibly

- Make shots sound/feel stronger
- Make flames/energy appear around weapon for emphasis
- Make enemies take more visible damage (knockback, effect, etc.)

Only show upgrades when holding "tab" key, doesn't need to be constantly visible

Keep emphasis on health and ammo above other HUD elements

Keep weapon selection readable, instantly recognizable

Burst 3:

Having more concrete separation between gameplay (during each round) and upgrade/weapon acquiring (between rounds)

- Creates opportunity for overhauled interfaces
- Does this risk making the game/gameplay feel incoherent?

What if weapon acquisition was more ingrained into HUD (rather than being a menu) and upgrades (which improve stats but do not always factor into moment-to-moment gameplay) are kept on their own screen?

Weapon equip has interaction prompt and a short blurb with relevant images/info

Place upgrades in parallel rather than on a grid

- Hide information of all except hovered selection

Remove option to “skip” weapon equip, just allow it to be performed

15.6 Final evaluation

15.6.1 Final evaluation session one

What do you expect to change in this HUD upon firing a weapon?

The ammo count at the bottom right. Two, very representative.

What will you expect to change upon taking damage?

Uh, in the bottom left. A two.

For eliminating an enemy, what would you expect to change here?

I would assume the round number in the top left in case that's the last one.

In this case, there's a progress bar that kind of fills up by one on the top left. How representative would you say that is?

Oh, I see it now. Uh, one, cause it was kind of hard to see the first one.

Upon advancing rounds, what would you expect to change?

The round count. Uh, two.

What would change upon charging up a shot, for example, if you could charge to hold a shot up.

The ammo count would go down by more than one shot. Uh, I think.

So this is what I'm proposing, um, to hold down a shot and kind of charge it up until you have, um, kind of a full like an icon that pops up.

Mm. I don't know if I would do it that way. Maybe just a horizontal bar into the crosshair. That's just how I've always seen charged shots be done though.

Okay. That's good to know. All right. So yeah, maybe approach it a different way. What would you rate this from negative to positive two?

Uh, I would say one.

What would you expect upon switching weapons?

Uh, the ammo count in the bottom left to change and also the name to change.

This is what I am kind of proposing: an icon change, as well as the model. What would you say for that?

A two.

What about gaining an upgrade? Where would you expect that to be?

Uh, I don't know. I guess it depends on your upgrades.

Okay. This is what I went with. Um, upon changing around and selecting an upgrade, this is sort of the change that you will see. What would you give for that?

Uh, um, I guess a two. Maybe like a 1.5.

—

This is my proposed upgrade screen. It's kind of like selecting, maybe like, perks from Black Ops Zombies. How well would you say these are in terms of representing the upgrades versus the system that I proposed before in the original play test?

Uh, they're better. They're miles better.

Is there anything in particular you appreciate about this or that you don't like about this?

The unique name and the description of what they do, the visual stat changes.

15.6.2 Final evaluation session two

I would like to see what you expect the changes will be upon firing the weapon.

Upon firing the weapon, I assume the bullet counter will go from 13 out of 13 to well above 13.

Okay, sweet. Yeah, so you were right on that one. And how representative would you say the HUD is of that behavior, from negative 2 to 2?

I'd say 2.

Okay. What changes upon taking damage?

The health in the bottom left goes from 150 to some other lower number.

Okay, perfect. This is kind of what I'm proposing. There's a change in health and also a visual indicator for direction. What would you give this one?

I'd say, also a 2, especially with the visual indicator. If it was just the number, it would probably be a 1, but with the visual it's a 2.

Okay, nice. What would change upon eliminating an enemy?

Well, the bullet would change because you have to shoot them presumably. Unless the enemy kills themselves somehow, I don't think that's possible. But the bullet will go down and if it's the last enemy in the round, it means the round will go back up?

Right, okay. Yeah, so what I'm proposing here, if it wasn't the last enemy of the round, that there would be a bullet change and also a little progress bar in the top left under the round counter. So that would increment. What would you give this one?

I'd say probably a 1 because I couldn't really tell, at least before an enemy was killed, that that progress bar was a progress bar. I just thought it was glare through the TV.

Right, okay. Nice. Now what would change upon advancing rounds knowing this?

Probably the round bar would go up and then the progress bar would go back down to 0.

Okay. Yeah, you're definitely right about that. Nice. What would you say for that?

Yeah, that's representative.

Okay. What would change upon charging up a shot of your weapon?

I assume that's what the little, is that what the little x-ray, lightning bolt meter under your health is, or is that somewhere else?

Okay, so this is what I was proposing, kind of a charging reticle that fills up as you charge. And there's a little green pop up when it's finished and ready to release. What would you give that one?

I'd say probably neutral because I didn't really know that you could charge up weapons, at least just from a hot air gun.

Right, okay. Okay, that's very helpful. What would change upon switching your weapon?

The weapon itself would change, presumably the bullet number would also change.

Okay. Yeah, so assuming the bullets stay the same... What would you say for that one?

I'd say it's two better because I like that it changes the type of ammo along with the actual weapon itself. I mean, it's very, very cute.

Nice, okay. And what would change upon gaining an upgrade?

Well, I mean, obviously depends on what upgrade you choose, but either the ammo goes up or the health goes up.

Okay, yeah, assuming you pick something up like other than that, something that would manipulate one of the numbers, the round would change, and you would sort of have this little icon at the bottom. What would you give this one?

I do like the icon, but at least for this one, I can't really tell what that icon, what that buff is supposed to be. Negative one.

Okay, so that was the first part. And just one other smaller part I was hoping to get. So, this is a prototype of the upgrade selection screen that I've created here. So this would sort of mimic the behavior of using the upgrade selection screen. How would you say this is either an improvement, or there are some elements that you don't like from the previous version. I'd like to hear your thoughts on this.

I think it's definitely an improvement. But one thing that may end up being a little weird is the fact that you kind of have to scroll through the whole thing.

Or can you just like hover your mouse? Or do you have to use the arrow keys?

The ideal would be hover your mouse. Or arrow keys.

I guess the one thing would be, you're not really sure what each upgrade is until you hover over.

Okay. That makes sense.

Like, you know, like for example, at this point it says, you know, "lemon lime," which is good, but, you know, I don't remember what the first, second, third, or any of the other milkshakes would be.

Right. Okay. That is really, really helpful. Sweet. Okay. Yeah, that's pretty much everything I needed to know. Do you have any other thoughts in general about what I've shown you, compared to the initial game?

I really like all the different stuff with the art you've been doing. It was really cool.

Thank you.

I'm glad the, I like the core identity of the game is starting to take shape more. I really think it's very impressive.

Thank you so much. Alright.

15.6.3 Final evaluation session three

So what would you think would change on the HUD when you're firing the weapon?

I mean, on the bottom right is your ammo, right. So I'd expect that number to decrease from 13 to 12. Right. Is this a single firing or is it a continuous action?

That's a good question. Yeah. So in this case, it would be just the number changing here because it would be like a single firing weapon. What would you give for this one in terms of how representative the HUD is of that action?

Yes, two.

Okay, sweet. What would you expect to change upon taking damage?

Depending on the quantity, I don't know what the quantity is of damage, but that bottom left looks like a health bar and I'd expect that number to go down. Maybe I see a flash of red on the corners of the screen. I don't know.

Okay, cool. Yeah, that's what I was kind of thinking we would do for the health. Sort of the number and a little visual directional indicator as well. What would you give for that one?

That's also a two.

Okay, sweet. What would change upon eliminating an enemy?

Okay, it depends on the kind of game we got here. But I mean, on the HUD, nothing necessarily needs to change, per se, unless there's a kill counter that appears somewhere, but I don't see any kill counter of any sort as is right now.

Okay, yeah, I'm glad you brought that up, because I am sort of trying to work in a kill, like a kill counter on the HUD with the little progress bar in the top left, it's very small. I recognize so that's just what I think that will definitely require some reworking. What would you give this one, score wise?

It's like a, it's like a zero. It's not obvious.

Okay, zero that makes sense. Thanks. What would change upon advancing rounds?

Probably that round number up there in the top left. That's about it, right?

Yeah. Sweet. Okay. What would you give for that one?

That's a two. Okay.

What would change upon charging your weapon? So this would be, there's a feature in the game where you can charge up a shot by holding your left mouse button.

Assuming that charging your weapon takes more ammo. I'd expect that number to decrease a larger amount on the bottom right. Maybe I'd see a red glow around the part of the HUD, a war on the weapon itself. Or, some sort of color change to indicate a charge being made or a movement of some sort.

Awesome. That's actually spot on— really helpful. We're thinking like a light, a lighting effect, as well as a sort of like icon that fills as you charge. When it's full, the little green stem of the cherry appears. This is based off of an upgrade I'm developing called Charge-Up Cherry. Okay. What would you give for this one?

That's a two.

Okay. Sweet. I'm surprised you guessed the light up. What changes upon switching weapons?

On that bottom right, it says pistol. Right? So I imagine the bullet type will change according to the weapon you change to. The ammo counter will change. Maybe the color of that bottom right portion will change. Obviously, the weapon you're holding changes, right? Yeah, that's my expectation.

Okay. Yeah. So definitely like the name change, the icon change, as well as the holding weapon change. There will be a different version for each of the weapons. What would you give this one?

That's a two.

Sweet. What would change upon gaining an upgrade?

How to upgrade for what?

It could be, for this instance, it'll just be an upgrade to how long you can charge your shots. So, nothing numerical on screen.

Okay. Okay. An upgrade to how long you can charge your shots. I mean, something's got to change with the weapon. Either that or when you charge up the shots, the color of the shot changes or maybe the lighting that appears around the crosshair looks a little bit different or more elongated or brighter, more vibrant. Something of that sort. But something to do with the weapon or the crosshair or a combination of both. So assuming that the upgrade doesn't change the ammo quantity, right? If it does change the ammo quantity, you'll want to reflect that in the number.

You actually did bring up a lot of things we're considering right now for the purpose of this. I would just be denoting like each different upgrade with a little icon at the bottom of the screen. But a lot of those effects would actually happen later on. I just didn't include them right here. So, good call. What would you give this one?

I wasn't expecting that, but I'll give it a one because it does make sense.

Okay, that's fair. Yeah. Mostly with your expectation. That's what I'm kind of gauging in the first part. And so, yeah, thank you. That was very helpful, actually. Okay, so that's

the first part here. And then just a really quick other element is the other smaller part. This is sort of the upgrade screen that I am proposing to kind of work through selecting the different upgrades between each round. I would just like to hear your thoughts on this sort of prototype here. Are there any glaring issues you see or things that you particularly like?

Okay, so you're shifting through all of them. Can you show me all of them again?

Yeah, of course. Regen Root Beer with health regeneration. Tanky Tamarind with maximum health increase. There's Stamina, a damage upgrade, increasing critical hit chance, charging your shots up for longer, and increasing your maximum clip size.

Okay, so I like it as it is so far. It's very good. I don't know if this poses a stylistic concern, but I like that there's a color difference for each type of upgrade. But if you... I would normally expect like a slightly altered icon as well for each class of upgrade, if that makes sense, because some of them are related to your gun. Some of them are related to your health or stamina. Some of them are related to reload speed and whatnot. But again, that's very subjective and not super streamlined in some ways. But otherwise, I like the upgrades screen here. Do these upgrades cost something or how does it work?

That's a good question. The idea would just be between each round, you would have the opportunity to select an upgrade, which some of them could be stacked. So maybe this round if you select it once, your maximum clip size will be 10 ammo, but the next time you get it, it might be 16 ammo for the increase.

But something I would also like to see actually, that is if you have gotten a certain type of upgrade in the previous round, make it so that the drink you're hovering over looks different, like bigger. For example, to indicate that you got it in the past and it kind of gives a nice little progression looking type system. It's an opportunity for some extra design, not that it's necessary.

15.6.4 Final evaluation session four

So, what would change upon firing the weapon?

This one. It would go down.

And how representative would you say my UI is able to communicate or represent that change in state?

Oh, um, very?

Um, what changes upon taking damage?

The pink thing.

Okay. Nice. So yeah, the number would go down, you have some directional stuff... How representative would you say that one is?

I wish it had something other than just a little charging thing, but I feel like it reflects it. So yeah, two.

That's good. Yeah. Both of those points are valuable. I'm going to take both of them into account. What would change upon eliminating an enemy?

Nothing.

So one of the things I'm planning to do is to increase this little progress bar up here at the top.

I didn't know that was a progress bar. And it's really small.

What would you rate that one then?

Uh, I would put it pretty low. Just because I wouldn't see it at all. These ones have big numbers that change that are like the size of these block things. So I'm like, oh great. Big things. Changing that, I wouldn't notice that.

What would you put this at on the scale?

Negative two.

Okay. What changes upon advancing rounds?

The round.

Nice. Okay.

That I've been waiting for.

What would you say for this one?

Very representative. This is the only one I was positive on as soon as you opened the screen. I was like, oh I know what that one does.

What would change upon charging the weapon? So this is a mechanic where you hold down the button to charge up a shot.

Oh, that's what the little tiny charge is? I thought that was part of the health bar.

Well, what would you think would change upon charging up a weapon?

This thing.

So, this is how I would represent a charged up shot, ready to fire. What would you rate that one here?

Well, I wouldn't anticipate it. But, uh, Like it's pretty clear I can fire the weapon, so I guess like one.

Okay, okay. But yeah, not something you would expect?

No, not at all. I was like, uh, I guess I misinterpreted what something else was. Well, because it came out of nowhere. There's no, like, outline for it.

What would change upon switching weapons?

What this thing looks like. I like that guy.

Would you say that this is representative or not representative?

Um, yeah, I think that one's fine.

Okay. What do you rate that number wise?

It's another one I wasn't anticipating, but once it happened I was like, oh, okay. Okay. So that's a one.

Um, and what would you expect to change upon getting an upgrade?

That like a number would increase?

That's true. Yeah, for a certain upgrade the number would increase. Um, and additionally I would add a little icon like this here. Um, what would you rate that one?

Negative two. I have no idea what that means.

Negative two. Okay. That's good to know.

Like there's a little tiny thing, but I just thought it was another target thingy.

Right. Okay. And then one other part here. Okay.

So this is a prototype I made of the upgrade screen between rounds.

Oh, I recognize that from the other upgrade. That just bumped up my other score I guess because I now know what that means. Um, but it was really small on the other screen. I didn't see it. Um, I really like how you do the letters. But I find these ones really hard to read. I don't know why. I'm like sitting here squinting going T-A-N-K-Y. There's too many mechanics. And they would... you'd populate a whole bunch of them on your screen when you get them? That's too clunky. But I like these being unique things, but they should go on the side, not on the bottom.

15.6.5 Final evaluation session five

What would change upon firing the weapon?

The bottom right. The bullet count.

Alright, yeah, that's what I was going for there. The bullet count would change. Sweet, and what would you give for this one on a scale of -2 to 2?

Two, absolutely. By the way, side note, I think the bullet is really cute and I like the fallout style you used there. That's really nice.

Thank you, that was what I was going for. What changes upon taking damage?

The health bar on the bottom left and I'm assuming the screen might go a little bit red but I don't know if that's what you're getting at because that's not HUD. So I guess, bottom left.

Alright, so yeah, you actually nailed both aspects of that. That's what I was going for, a little bit of a directional damage thing to see where you're getting hit from, as well as the health value changing. What would you give that one?

That's a 2.

What would change upon eliminating an enemy?

I guess it depends. Well, it'd be the bullet in the bottom right, assuming you fired your weapon. If it was the last enemy, then the top left, the round might change. Other than that, nothing, I'm guessing.

Okay, yeah. You're definitely right on both of those. Some people have had difficulty seeing it, which is definitely reflecting on my design. But at the top left, there's a little bit of a progress bar that changes as you kill one. But if it was the last one of the round, yes, the round number would definitely change there. So nicely done. What would you give that one?

I feel like I'd give that a 1 because it does show the thing, but yeah, I feel like that'd be more of like, if someone really likes a game, they go like, oh dude, I didn't even notice it. That's, but that's sick. But it is, it's like not overt. I guess that is the way I'd put it.

Thank you. What would change upon advancing rounds here?

The number. Top left. And the progress bar would reset.

Yes. What would you give that one?

That's a 2.

What about charging up your weapons? So to explain, there's a mechanic where you can hold down the mouse button to hold your shot and charge it up to increase its power. What would you expect to happen there?

I'd expect the yellow things on the gun to potentially change color. Maybe some shaking. Maybe the screen zooms in a little bit sometimes. And at least after you shoot it in the bullet counter kind of goes down or, I don't know that actually might not if it's a different meter. But yeah, at least I expect that the gun to be where most of the visuals are happening.

So this is what I was kind of going for. There would sort of be an effect like this bar fills up and then once it's done, it has this little thing on top of here. For context. It's matching with an ability called charge up cherry. So that's why it kind of looks like a cherry. What would you give for this one?

Would you mind going back so I can see what it was like before you changed it? In that case, I feel like I'd give it a one because once you're charging, I feel like that does show what you're going for. But yeah, I guess I just like, I don't know, sometimes like in shooters, like you have like an empty bar and then when you start charging, it fills up. You know what I'm saying?

Yeah. That's definitely something I've been considering. And I definitely agree. It's maybe not the most discoverable ability right now.

When you're using it, it is apparent that you're using some kind of special shot.

What would you expect to change upon switching weapons or weapon type?

The gun would be different and the bottom right part of the hub would be different.

Okay. Yeah, this is what I was going for. There's a weapon in the game where it's essentially a modified microwave. So the name and the little character would change as well.

You change the character, dude. That's tip top. I love that. That's a two. That's a three if I could say it.

What would you expect to change upon gaining an upgrade to one of your abilities?

Are there abilities other than the charge of shot?

There are. Yes.

Oh, well, um, maybe like a, I could expect like an icon to appear maybe in the top right, like a perk, or some visible change to the gun or depending what upgrade is like some kind of change to either the health or the bullets.

Okay, yeah, you're pretty much right on all of those points. Like, yeah, the numbers would definitely increase for certain upgrades. And I was sort of thinking like a little perk icon on iconography. Position, definitely up for discussion. What would you give that one?

Yeah, that's very representative.

The upgrade selection screen that I designed here is another little bit of the prototype. I would love to hear your thoughts on this one, especially how it compared to the screen from before.

Okay, well, no shade to the person who did the original HUD. But the difference between that and this is like the first one might as well be like a bullet point list. Like, considering that cherry and whole food thing. This looks awesome. It looks tasty. It's fitting with the whole pip-boy aesthetic. Yeah, dude. I love it.

All right. I'm really glad to hear that. Sweet. And yeah, if you have any, any feedback or comments in general, I just really love to hear it.

I don't know if this is specifically what you'd be working on. But like I very specifically thought, like with the gun with the yellow things on it, I feel like that's a good spot to show some kind of like ability where like if the yellow thing changes, I feel like that's a cool spot.

15.6.6 Quantitative results

Table 8—Descriptive statistics arising from quantitative analysis.

	Mean	Standard Deviation	Interquartile Range
Firing Weapon	2	0	0
Taking Damage	2	0	0
Eliminating Enemy	0.2	1.303840	1
Advancing Rounds	2	0	0
Charging Weapons	1	0.707106	0.25
Switching Weapons	1.8	0.447213	0
Obtaining Upgrade	0.4	1.816590	2.25

15.6.7 Coded qualitative results

Table 9—Coded results organized by subject.

1 - Firing Weapon	
Bottom right, ammo count decreasing	5

2 - Taking Damage	
Bottom Left	5
Flash of Red	2

3 - Eliminating Enemy	
Round Number Increase	3
Kill Counter	1
Bullet Change	2

4 - Advancing Rounds	
Round Count	5
Reset Progress Bar	2

5 - Charging Weapons	
Ammo Count Decrease	3
Uses More than One Ammo	2
Charge/Stamina Meter	2
Glow	1
Color Change	2
Screen Effects/Shaking	1

6 - Switching Weapons	
Ammo Count Change	3
Weapon Name Change	3
Icon Change	2

7 - Obtaining Upgrade	
Unsure	1
Numbers Increasing	4
Depends on Selection	3
Stylistic Change	2
Icon	1